

Yeji Lee

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Portfolio (website): <https://yejilee2305.github.io/>

EDUCATION

Michigan State University

Bachelor of Science in Computer Science, Minor in Business

East Lansing, Michigan

2022 - 2025

- Dean's List
- Relevant Coursework: Algorithms & Data Structures, Computer Architecture, Software Design, Artificial Intelligence, Database Systems, Mobile App Development

EXPERIENCE

LG Energy Solution Michigan

Vision Systems Engineer I, Lamination & Stacking

Lansing, MI

February 2026 – Present

- Support L&S inspection equipment during battery cell production, including daily monitoring, issue follow-up, and communication with production, quality, and equipment teams.
- Worked through early production stages, including stress run and start of production (SOP) / mass production ramp-up, helping stabilize inspection equipment and follow up on open issues.
- Cover day-shift inspection support while coordinating with the night shift engineer and senior engineer to keep issue tracking and handoffs aligned.
- Support deep learning image inspection work using internal tools, including image review, model training/application support, and inspection result monitoring.
- Assist troubleshooting for false NG, defect detection issues, equipment abnormalities, and process-related inspection trends.
- Participate in PM/BM activities by following up on equipment issues, technician requests, abnormality reports, and recurrence prevention items.
- Use the CS background to approach inspection issues through structured troubleshooting, data review, and clear technical communication.

Vectra AI (via Michigan State University Capstone)

Software Engineer

East Lansing, Michigan

August 2025 – December 2025

- Built a distributed FastAPI orchestration pipeline for RFC discovery, YAML extraction, PCAP generation, and multi-stage validation, reducing manual integration time by 45%.
- Developed a React dashboard with FastAPI endpoints to visualize real-time system metrics backed by SQLite and file-based job orchestration.
- Designed an LLM-powered semantic validation pipeline that compared user intent with Vectra AI metadata and generated deterministic validation checks per PCAP.
- Implemented an RFC web crawler that scraped and normalized 35+ RFCs across multiple mirrors, enabling automated protocol discovery and freshness validation.
- Designed multi-packet flow and TCP variant generators producing 3–12 packet exchanges per session, validated with Wireshark dissectors for protocol conformance.
- Collaborated with stakeholders to translate protocol and validation requirements into structured Python pipelines.

Formula SAE, Michigan State University

Mechanical / Software Design Member

East Lansing, Michigan

August 2022 - May 2023

- Designed and built a custom tire rack for the race car that improved pit garage space by 20% for pit stops.
- Collaborated with 10+ engineers to design CAD models, manufacture components, and assemble race-ready systems.
- Used simulation and validation checks to review the weight distribution, stability, and durability of the design.

PROJECTS

Packet Forge: AI Network Protocol Engine

- Built a natural-language-driven PCAP generation system that translated user intent into RFC-compliant packet flows.
- Built a CLI tool integrating three RFC mirror sources, normalizing 80+ protocol aliases and mapping 35+ network protocols to official RFCs.
- Developed an automated multi-packet flow generator producing 5–12 stateful packet flows per session, validated using Wireshark dissectors.
- Implemented TCP transport variant generators to produce test cases for protocol robustness testing.
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Automated Meeting Notes System

- Built an AI-powered app processing 60+ minute audio/video files, reducing manual note-taking time by 90%.
- Integrated OpenAI Whisper and GPT-4 to support transcription, action item extraction, and automated priority tagging.
- Developed a FastAPI backend with PostgreSQL database and Slack/email notifications for downstream workflow updates.

SKILLS

Manufacturing & Inspection: EV Battery Manufacturing, Lamination & Stacking, Inspection Systems, Machine Vision, Deep Learning Image Inspection, Image Review, Model Training Support, Inspection Result Monitoring, False NG Monitoring, Defect Detection, Equipment Troubleshooting, PM/BM Support, Root Cause Analysis

Programming: Python, C++, Java, JavaScript, TypeScript, SQL

Frameworks & Backend: FastAPI, React, Node.js, REST APIs

Systems / Networking: PCAP Analysis, Wireshark, Network Protocols, RFC Parsing

Cloud & DevOps: AWS, Docker, Linux, Git